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Lab 4 Task 1: A by hand (g, h, f)

Given graph (travel-time units), start = S, goal = G

Edges: S → A = 1, S → B = 4, A → C = 2, B → G = 5, C → G = 3

Heuristich: S = 5, A = 4, B = 4, C = 2, G = 0

Formulas: g(next) = g(node) + edge cost, f = g * h

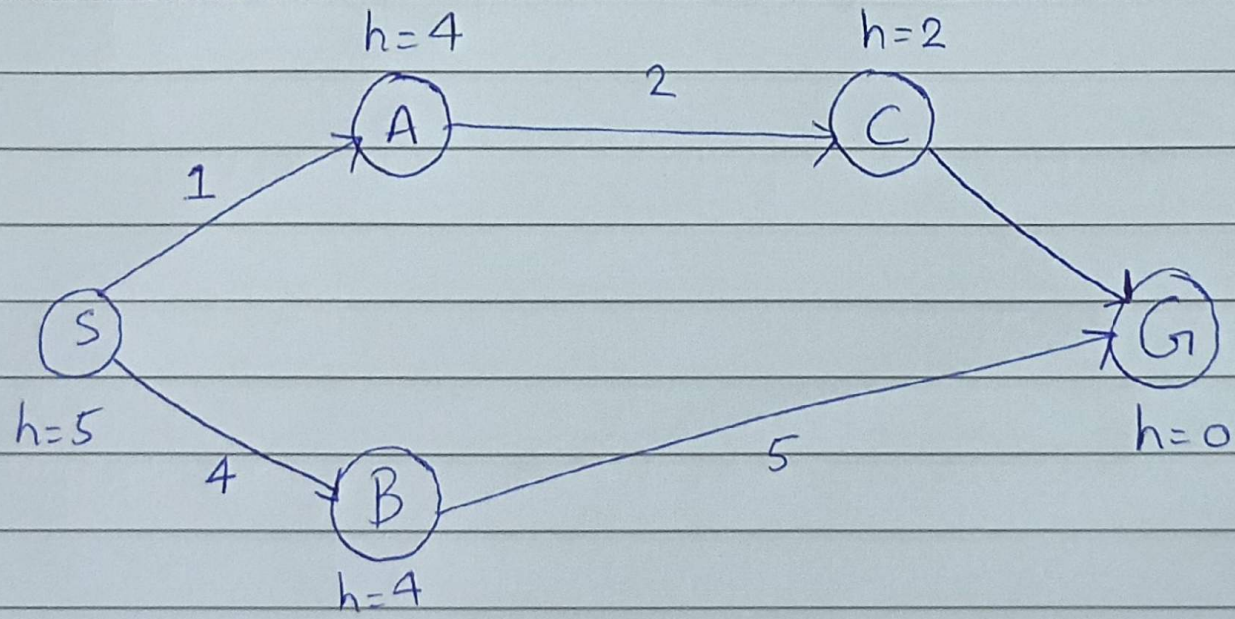
Rules used every time (same as the code):

1) always pop the smallest from the frontier

2) If f ties → smaller g first, then alphabetical name.

3) Successors are checked in the order listed above.

4) Node counts as expanded when successors are examined.



(red = edge cost, green = h)

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True remaining cost h^*: S=6, A=5, B=5, C=3, G=0

$h <= h^*$ for every node ($5 <= 6, 4 <= 5, 4 <= 5, 2 <= 3$) → admissible

Consistency $h(u) <= cost(u,v) + h(v)$:

S→A $5 <= 1+4 = 5$ ok S→B $5 <= 4+4 = 8$ ok

A→C $4 <= 2+2 = 4$ ok B→G $4 <= 5+0 = 5$ ok

C→G $2 <= 3+0 = 3$ ok all pass

Start: frontier = {S (g=0, h=5, f=5)}

Expansion 1: select S

$g(S) = 0, h(S) = 5, f(S) = 0+5 = 5$

Successors of S in order:

A: $g = 0+1 = 1$, $h = 4$, $f = 1+4 = 5$

B: $g = 0+4 = 4$, $h = 4$, $f = 4+4 = 8$

Frontier now:

A(g=1, h=4, f=5) B(g=4, h=4, f=8)

Next: f(A)=5 is smaller than f(B)=8 → pick A

Expansion 2: Select A

$g(A) = 1, h(A) = 4, f(A) = 1+4 = 5$

Successor of A:

C: $g = 1+2 = 3$, $h = 2$, $f = 3+2 = 5$

Frontier now:

C(g=3, h=2, f=5) B(g=4, h=4, f=8)

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Next: $f(C) = 5$ is smaller than $f(B) = 8 \rightarrow$ pick C

Note: f stayed 5 on S, A, C. The heuristic is consistent, so f never decreases along a path. No f-tie happened, so the tie-break rule (g, then name) was not needed here.

Expansion 3: Select C

$g(C) = 3,\ h(C) = 2,\ f(C) = 3 + 2 = 5$

Successor of C:

$G_1: g = 3 + 3 = 6,\ h = 0,\ f = 6 + 0 = 6$

Frontier now:

$G_1(g = 6, h = 0, f = 6)$ $B(g = 4, h = 4, f = 8)$

Next: $f(G_1) = 6$ is smaller than $f(B) = 8 \rightarrow$ pick G_1

G_1 is the goal → stop (goal is logged, not counted as an expansion). B was never expanded.

Summary of the first three expansions

Step	node	g	h	f	frontier left after it
1	S	0	5	5	A(f5), B(f8)
2	A	1	4	5	C(f5), B(f8)
3	C	3	2	5	G_1(f6), B(f8)

Result: path $S \rightarrow A \rightarrow C \rightarrow G_1$

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Cost check: 1 + 2 + 3 = 6 (other route S - B - G = 4 + 5 = 9)

A* expanded 3 nodes : S, A, C

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Task 2: A* vs UCS (same graph, same counting rule)

UCS = same code with every h = 0, so f = g

A* traces: S, A, C, then G (goal) -> expanded = 3

UCS trace: S(g0), A(g1), C(g3), B(g4), then G(g6)

UCS expanded 4: B (g=4) is cheaper than G (g=6) so it is opened first; B -> G gives 4+5=9, not better than 6, no push.

Both return S-A-C-G, cost 1 + 2 + 3 = 6 (summed independently)

stale entries skipped: 0 in both runs. Goal not counted.

A* saved 1 expansion (3 vs 4): it never opened B.

Task 3: Overestimate h(C): 2 -> 10

True remaining cost of C = 3, so h(C) = 10 overestimate by 7.

S: g0 h5 f5 -> push A f = 1 + 4 = 5, B-f = 4+4 = 8

A: g1 h4 f5 -> push C f = 3 + 10 = 13 (C looks bad now)

Frontier: B (f8), C (f 13) -> pick B, expand B

B: g4 h4 f8 -> push G g = 4+5 = 9, f = 9

Frontier: G (f9), C (f 13) -> pick G, goal reached

Result: S -> B -> G, cost 9 (was 6), expanded S, A

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- Path and cost changed; count stayed 3 (but C was skipped)

Why: the big h(C) pushed the good route behind G (f = 9), so the worst goal was popped first. An overestimate breaks the optimality guarantee; one unchanged run would not prove a heuristic safe, so always check $h <= h^*$ on every node.